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MILITARY SPECIFICATION

CLOTH, LAMINATED, NYLON TRICOT KNIT, POLYURETHANE FOAM

LAMINATE, CHEMICAL PROTECTIVE AND FLAME RESISTANT

This specification is approved for use by Natick Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers three types of tricot knit nylon cloth laminated to polyurethane foam and impregnated with an activated carbon mixture.

1.2 Classification. The cloth shall be of the following types as specified (see 6.2).

Type I -Chemical protective
Type II -Chemical protective and flame resistant
Type III -Chemical protective (special finish)

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATION

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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FEDERAL

PPP-P-1136 - Packaging of Coated (Plastic; Rubber) and Laminated Fabrics

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3574 - Flexible Cellular Materials, Slab, Bonded, and Molded Urethane Foams

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials.

3.2.1 Base cloth. The base cloth shall be tricot knit nylon knitted on a 28 gauge tricot machine using 40 denier nylon yarn. The cloth shall

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approximate color Olive Green 106 or black, and shall show "good" fastness to water. The cloth shall also meet the physical requirements specified in table I when tested as specified in 4.4.1.1.

TABLE I. Physical requirements of base cloth

Characteristic	Requirement Minimum	Maximum
Weight, ounce/square yard	1.8	2.2
Yarns per inch:		
Wales/inch	37	-
Courses/inch	35	-
Bursting strength, pounds:	75	-

3.2.2 Polyurethane foam. The foam to be adhered to the base cloth shall be an unpigmented polyester based polyurethane flame laminable foam meeting the physical requirements in table II when tested as specified in 4.4.1.1.

TABLE II. Physical requirements of polyurethane foam

Characteristic	Requirement Minimum	Maximum
Thickness, inches	0.089	0.109
Density, pounds/cubic foot	-	2.0
Tear, pounds/inch	3.0	-
Compression set, percent	-	25.0

3.2.3 Impregnating mixtures.

3.2.3.1 Type I. The impregnating mixture for type I cloth shall be a carbon slurry consisting of activated carbon in water with other ingredients to bind the carbon to the polyurethane (see 6.4 and 6.4.1.1).

3.2.3.2 Type II. The impregnating mixture for type II cloth shall consist of activated carbon and 4 percent Phos Chek^{R} P-30 in water with other ingredients to bind the carbon to the polyurethane (see 6.4 and 6.4.1.2).

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3.2.3.3 Type III. The impregnating mixture for type III cloth shall consist of activated carbon (see 6.4 and 6.4.1.3) and Polyoz WSRN-80^R and Phos Check^R P-30 in water with other ingredients to bind the carbon to the polyurethane. The impregnating compound shall be made up of formulations A and B, mixed together in the quantities specified in formulation C to attain an approved impregnation compound. The final composition (formulation C) shall be applied to the cloth within 48 hours after preparation (see 4.4.1.2).

Impregnating mixture for type III cloth

Formulation A 1/	Percent (by wt.)
Activated carbon	35.0
Dispersing agent (18% solids)	5.0
Water	29.0
Thickener (28% solids) 5% solution	25.0
Binder (38% solids)	6.0
	100.0

Formulation B 2/	Percent (by wt.)
Polyox WSRN-80^R 3/	4.5
Phos Chek^R P-30 3/	36.0
Water	59.5
	100.0

Formulation C 4/	Percent (by wt.)	Wt. (lbs.)
Formulation A (Carbon slurry - 40% solids)	70.0	315
Formulation B (Polyox WSRN-80^R and Phos Chek^R P-30)	11.0	50
Water	19.0 5/	85
	100.0	450

1/ Carbon slurry (see 6.4.1).

2/ Preparation of formulation B. The Polyox WSRN-80^R solution shall be prepared first and the Phos Chek^R P-30 added to the Polyox WSRN-80^R suspension. The required amount of Polyox WSRN-80^R shall be placed in 2/3 the amount of required water with high speed stirring to dissolve the Polyox WSRN-80^R completely. The Phos Chek^R P-30 and the remaining water shall then be added slowly with high speed stirring to disperse the Phos Chek^R P-30 in the Polyox WSRN-80^R solution.

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3/ See 6.4.1.6.

4/ Preparation of formulation C. Small quantities of formulation B and water shall be added simultaneously into formulation A with high speed stirring to insure a uniform dispersion. Batch size may vary at the contractor's option.

5/ The amount of water may be decreased as needed to obtain the required laminated cloth weight.

3.3 Assembly and treatment of the cloth.

3.3.1 Bonding of base cloth to foam. Prior to impregnation, the polyurethane foam shall be flame laminated to the float side of the nylon knit base cloth. The cloth/foam laminate shall meet the following flexibility and thickness requirements when tested as specified in 4.4.1.1.

	Requirement	
Flexibility:	Minimum	Maximum
Wales, load pounds	-	0.010
Courses, load pounds	-	0.010
Thickness (inches)	0.070	0.094

3.3.2 Impregnation. When type I, II, or III is specified, the impregnating compound specified in 3.2.3.1, 3.2.3.2, or 3.2.3.3, as applicable, shall be applied to the laminated cloth by a pad-dry procedure. The padding of the dispersion shall be done at room temperature by passing the cloth through squeeze rolls and then drying the cloth (see 6.4.2). Control minimum shrinkage by overfeed prior to impregnation and minimum pin frame tension during oven drying (see 6.4.2).

3.4 Finished cloth. The finished cloth (impregnated cloth/foam laminate) shall meet all the physical requirements in table III when tested as specified in 4.4.3.

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TABLE III. Physical requirements of finished cloth

Characteristic	Requirement		
	Type I	Type II	Type III
Weight, ounces/square yard:			
Minimum	7.0	9.0	7.5
Maximum	9.5	10.0	10.0
Thickness, inches:			
Minimum	0.070	0.090	0.070
Maximum	0.110	0.120	0.120
Air permeability, cubic feet/minute/square foot (minimum)	40	40	40
Bursting strength, pounds (minimum)	75	75	75
Carbon tetrachloride absorption, mg/cm^2 (minimum)	1.8 1/	1.2 2/	1.8 1/
Flexibility-load pounds (maximum) 60° angle:			
Wales	0.050	0.065	0.065
Courses	0.030	0.045	0.045
Flame resistance, vertical:			
After flame, wales and courses direction, seconds (maximum)	-	4.0	-
After glow, wales and courses direction, seconds (maximum)	-	6.0	-
Char length, wales and courses direction, inches (maximum)	-	6.0	-
Water resistance, mL (maximum)	-	-	50 3/

1/ No individual specimen test result shall be less than 1.6 mg/cm^2.

2/ No individual specimen test result shall be less than 1.0 mg/cm^2.

3/ Maximum amount of water penetrating the sample within a 5 minute period.

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3.5 Length and put-up. The finished cloth shall be put up in rolls 35 to 110 yards in length. No piece in a roll shall be less than 10 yards in length. No more than one splice per roll is permitted.

3.6 Width. The width of the finished cloth between clip marks shall be 50 to 60 inches.

3.7 Surveillance marking.

3.7.1 Lot number marking. Each roll from each lot of finished cloth shall be identified with a specific lot number consisting of the following code in the designated format:

AB 1/ - CDOEF 2/ - 000000 3/

1/ First two letters of the fabric impregnator's company name, using capital letters.

2/ Applies to type III cloth only.

A. First two letters of the formulation "C" mixer's company name.
B. The middle zero shall designate the daily batch number (1-9).
C. First two letters of the formulation "B" mixer's company name.

3/ Date

A. First two zeros shall indicate day of fabric impregnation.
B. Middle two zeros shall indicate month of fabric impregnation.
C. Last two zeros shall indicate year of fabric impregnation.

Labels on which the lot number is marked shall be securely attached to each roll of impregnated laminated cloth.

3.7.2 Lotting. The lot of finished cloth is defined as the number of yards of laminated cloth impregnated in no more than one day. For type III cloth, the lot shall also consist of only that cloth impregnated using one batch of formulation "C" and formulation "C" shall only contain one mixture of formulation "B".

3.8 Color. The color of the finished cloth shall be a suitable shade of black.

3.9 Workmanship. The finished cloth shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2.1 and 4.4.2.2, and tested for the characteristics specified in 4.4.3. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Component testing. In addition to the quality assurance provisions of applicable subsidiary documents, testing shall be performed on components and materials listed in table V for the characteristics shown. Test methods cited shall be in accordance with FED-STD-191 except where otherwise specified. The sample units for the base cloth, the polyurethane foam, and the cloth/foam laminate (prior to impregnation) shall be 3 continuous yards each full width. The sample size (number of sample units) shall be as specified in table IV. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. All test reports shall contain the individual values utilized in expressing the final results.

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TABLE IV. Sample size

Lot size (yards)	Sample size (No. of sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and more	5

TABLE V. Component tests

Component	Characteristic	Rqmt. para.	Test method	No. deter. per sample unit	Results reported as
Base cloth	Knit	3.2.1	1/	-	-
	Gauge	3.2.1	1/	-	-
	Yarn denier	3.2.1	1/	-	-
	Colorfastness to water	3.2.1	5630	1	Pass or fail
	Weight	3.2.1	5041	5	Nearest 0.1 oz.
	Yarns per inch	3.2.1	5070	5 each direction	Whole number
	Bursting strength	3.2.1	5120	5	Nearest 1 lb.
Polyurethane foam	Composition	3.2.2	1/	-	-
	Thickness	3.2.2	ASTM D-3574 2/	3	Nearest 0.001 in.
	Density	3.2.2	ASTM D-3574	1	Nearest 0.1 lb./ft.^{3}
	Tear	3.2.2	ASTM D-3574 2/	3	Nearest 0.1 lb./in.
	Compression set	3.2.2	ASTM D-3574 3/	3	Nearest 0.1 percent
Impregnating compound	Compositon	3.2.3.1, 3.2.3.2, and 3.2.3.3	1/	-	-

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TABLE V. Component tests (cont'd)

Component	Characteristic	Rqmt. para.	Test method	No. deter. per sample unit	Results reported as
Cloth/foam laminate (prior to impregnation)	Flexibility	3.3.1	5202	5 each direction	Nearest 0.001 pound
	Thickness	3.3.1	4.5.1	5	Nearest 0.001 inch

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ Except that the test specimen shall be a piece of foam 2 inches by 6 inches in the thickness supplied. A slit 2 inches long shall be cut in one end of the specimen parallel to the long dimension to form two 1-inch wide legs which shall be gripped in opposite jaws of the testing machine. Thickness shall be measured with a dial gage with a 20-square inch foot area (minimum) and a foot pressure of 0.010 pounds per square inch.

3/ Deflection shall be 50 ± 1 percent of the thickness of the specimen and compression set shall be calculated as percentage of the original thickness.

4.4.1.2 Certification (type III). A certificate of compliance shall be furnished with each lot of cloth stating that the cloth was processed or finished with the impregnating compound (formulation C) within 48 hours after the compound was prepared (see 3.2.3.3).

4.4.2 End item examination.

4.4.2.1 Yard-by-yard examination. The required yardage of each roll shall be examined on both sides for the defects listed below. All defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition, in which case only the more serious defect shall be counted. The sample unit shall be 1 linear yard. The acceptable quality level (AQL), expressed in terms of defects per hundred yards, shall be 2.5 for major defects and 6.5 total (major and minor combined) defects . The inspection level shall be II. The number of rolls from which the sample yardage is to be selected shall be in accordance with table VI. The sample yardage shall be apportioned equally among the selected rolls.

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Defect	Classification Major	Classification Minor
Cloth side		
Color not as specified		X
Hole, cut, tear, mend, run or dropped stitch, or abrasion mark resulting in weak area		
- from 1/2 inch to 1.0 inch in any direction		X
- larger than 1 inch in any direction	X	
Creases or area of delamination	X	
Foreign matter		X
Oil spot or stain		X
Objectionable odor	X	
Polyurethane foam side		
Cut, tear, void, or repaired section		
- 1/2 inch or less in diameter		X
- over 1/2 inch in diameter	X	
Foreign matter	X	
Rough or ragged edges (other than tenter clip marks)	X	
Defective foam splice	X	
Tackiness	X	
Uneven distribution of impregnating compound	X	
Polyurethane foam does not cover the entire width of the knit backing cloth between tenter clip marks	X	

TABLE VI. Sample size

Lot size in yards	Sample size in rolls	Acceptance number 2/
Up to and including 1200 1/	3	0
1201 to and including 3200	5	0
3201 to and including 10,000	8	0
10,001 to and including 35,000	13	0
35,001 to and including 150,000	20	1
150,001 and over	32	2

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

2/ Applicable to length examination defects only (see 4.4.2.2).

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4.4.2.2 Length examination. Each roll of cloth used in the yard by yard examination shall be examined for the defects listed below. If the total number of defects in the sample rolls exceeds the applicable acceptance number specified in table VI or if the total of the actual lengths of the sample rolls is less than the total of the lengths marked on the roll tickets, the lot shall be rejected.

Any roll (gross length) less than 35 or more than 110 yards.
Any roll (gross length) more than 2 yards less than gross length marked on ticket.
Any piece less than 10 yards.
More than one splice per roll.
Any roll containing 2 or more pieces unspliced.
Width less than 50 inches or more than 60 inches (between clip marks).

4.4.3 End item testing. The laminated cloth shall be tested for the characteristics indicated in table VII. The methods of testing specified in FED-STD-191 wherever applicable and as listed in table VII shall be followed. The sample unit shall be 3 yards full width of finished cloth. The sample size shall be in accordance with table IV. The lot shall be unacceptable if the lot average for air permeability, if any sample unit test result for any other characteristic, or if any individual carbon tetrachloride specimen result fails to meet the specified requirements. All test reports shall contain the individual values utilized in expressing the final results.

TABLE VII. End item tests

Characteristics	Requirement paragraph	Test method	No. determinations per sample unit	Report results to nearest
Weight	Table III	5041	5	0.1 oz./yd.^{2}
Thickness	Table III	4.5.1	5	0.001 inch
Air permeability	Table III	5450 1/	5	0.1 ft.^{3}/min/ft.^{2}
Bursting strengh	Table III	5120 2/	5	1 pound
Carbon tetrachloride absorption	Table III	4.5.2 3/	9	0.1 mg/cm^2

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TABLE VII. End item tests (cont'd)

Characteristics	Requirement paragraph	Test method	No. determinations per sample unit 1/	Report results to nearest
Flexibility	Table III	5202	5	0.001 load pounds
Flame resistance, verticle:				
After flame (type II only)	Table III	5903	5	0.1 second
After glow (type II only)	Table III	5903	5	0.1 second
Char length (type II only)	Table III	5903	5	0.1 inch
Water resistance (type III only)	Table III	5516 4/	5	1 mL

1/ Report the lot average.

2/ The bursting strength tests shall be made with the specimen placed in the ring clamp nylon tricot knit side up with the foam side of the laminate facing the ball.

3/ Testing shall be performed by the Government. Except in cases of dispute, test samples shall be from production dried materials. In case of dispute, samples shall be conditioned in an air circulating oven at 50° ± 2°C for 16 ± 0.5 hours and tested while in the conditioned state.

4/ Prior to testing, adjust the flow of water so that 800 ± 50 mL will be delivered during a 5-minute test period with no sample in the apparatus. Samples shall be tested foam side up, the hydrostatic head shall be 5 cm and the sample shall be tested for 5 minutes. Water penetrating through the sample shall be collected and measured. Dry upper and lower orifice plate thoroughly between test samples.

4.4.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1136.

4.5 Methods of inspection.

4.5.1 Thickness measurement.

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4.5.1.1 Test specimen. The specimen shall be a piece of laminated cloth at least as large as the presser foot of the thickness gauge and shall be free of folds, creases, knots or other distortions which are not representative of the material surface. No selvage or cut edge shall be included in the sample tested. The specimens shall be cut from different parts of the full width of the laminated cloth.

4.5.1.2 Apparatus. The test apparatus shall be a compressometer with a 20 square inch presser foot (see 6.5).

4.5.1.3 Procedure. The compressometer shall be balanced. After balancing, the test specimen shall be placed under the presser foot and the machine set for 0.01 pounds per square inch. The presser foot shall then be lowered and the thickness reading under the above specified pressure shall be recorded.

4.5.1.4 Report. Five specimens shall be tested from each sample unit. The result shall be reported as the average of the five specimens tested.

4.5.2 Carbon tetrachloride absorption.

4.5.2.1 Apparatus and reagents.

4.5.2.1.1 Vapor generator. Any apparatus can be used that generates a carbon tetrachloride/nitrogen gas mixture which contains a concentration of 5 $\pm$ 0.5 mg of carbon tetrachloride per liter (see 6.7).

4.5.2.1.2 Diluent. Prepurified nitrogen shall be used as the diluent to pass the carbon tetrachloride vapors through the test cups containing the test specimens. The flow through each test cup shall be 1 $\pm$ 0.03 liter/minute. The rate of flow can be controlled by the use of either a rotometer or a critical orifice.

4.5.2.1.3 Constant temperature cabinet. The constant temperature cabinet shall be large enough to hold the system, as shown in figure 1, and shall be capable of maintaining a temperature of 90° $\pm$ 2°F.

4.5.2.1.4 Fabric sample cup. The fabric sample cup shall conform to the requirements shown in figure 2.

4.5.2.1.5 Pyrolyzer. The pyrolyzer shall conform to the requirements shown in figure 3. The platinum pyrolyzer is used between the outlet of the sample cup and the inlet of the "break-point" indicator bubbler. The pyrolyzer is heated by applying 10 volts, A.C., across the terminals during the test.

4.5.2.1.6 Bubbler. The bubbler shall conform to the requirements shown in figure 4.

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4.5.2.1.7 Schwartz-type drying tube with stop-cocks. The tube shall be filled with activated charcoal as specified in 4.5.2.1.8, loosely packed, as shown in figure 5. Glass wool, loosely packed, shall be placed on top of the activated carbon. The filled tubes shall be stored in a desiccator when not in use.

4.5.2.1.8 Charcoal, cocoanut, activated, 6 to 14 mesh.

4.5.2.1.9 Indicator solution. Dissolve 2.5 grams of potasium iodide in 1000-mL of deionized water. Mix 90-mL of this solution with 10-mL of a 1-percent, stable, starch solution. The starch/iodide solution is used in the indicator bubbler. Prepare a fresh starch/iodide solution daily.

4.5.2.1.10 Desiccator.

4.5.2.1.11 Analytical balance.

4.5.2.1.12 "C" type clamp.

4.5.2.1.13 Ice water bath.

4.5.2.2 Determination of carbon tetrachloride concentration.

4.5.2.2.1 Tare weight of drying tube. The Schwartz-type drying tube, filled with charcoal and prepared as shown in figure 5, shall be equilibrated to constant weight using the apparatus with a flow of 1 $\pm$ 0.03 liter/minute of nitrogen (see figure 1), no carbon tetrachloride. The Schwartz-type drying tube replaces the sample cup for this determination.

4.5.2.2.2 Determination of carbon tetrachloride concentration. Sample the carbon tetrachloride vapor from the generator manifold at a rate of 1 liter/minute through the tared Schwartz-type drying tubes (with stop-cocks) for a 15-minute period. Close stop-cocks, cut off the vacuum and disconnect the Schwartz-type drying tube. Allow the tube to come to thermal equilibrium for 10 minutes at ambient temperature. Open and close one stop-cock slowly to equilibrate pressure. Weigh the tube to the nearest milligram.

4.5.2.2.3 Calculation. The carbon tetrachloride concentration (mg/liter), is determined by dividing the weight pickup (in mg) by the sampling time (minutes), and flow rate (liters/minute). Determine the carbon tetrachloride concentration during each test run.

4.5.2.3 Test specimen. The test specimen shall be a 13.6 $\pm$ 0.2 cm diameter piece of the finished fabric. For each 3-yard sample unit cut three specimens equally spaced along left side, three specimens equally spaced along right side, and three specimens equally spaced along the center of the sample unit. Each specimen shall be tested.

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4.5.2.4 Procedure. Assemble the complete apparatus as shown in figure 1. For each test, mount one layer of the material in the test cup, foam side up. The up-side of the cup is the inlet side in respect to carbon tetrachloride flow. After placing the samples in the cup, clamp the cup halves together with a "C" type clamp. The clamp shall remain on the cup during the entire test. Place 10-mL of the indicator solution in each of the indicator bubblers and immerse the indicator bubblers in an ice water bath and keep immersed during the test. A white background shall be used behind the bubbler to facilitate color detection. Start the flow of carbon tetrachloride through the test sample and record the time for the first tinge of blue to appear in the indicator bubblers. This establishes the original recorded time as the true "break" time. This bubbler is replaced with a fresh bubbler which is run until the latest new bubbler "breaks" within a 2-minute period. If the second bubbler does not show the blue tinge, the test is continued until the second bubbler does show the blue tinge, and this additional time is added to the original break time to establish a new true "break" time. This bubbler is replaced with a fresh bubbler which is run until the latest bubbler "breaks" within a 2-minute period. The additional bubblers are required to confirm that a true "break" has been obtained. The sample cup shall be purged with air after running each sample to assure removal of any residual carbon tetrachloride.

4.5.2.5 Calculation. The carbon tetrachloride absorption shall be calculated as follows:

$$mg/cm^2 = \frac{T \times C}{A} \times F$$

Where: T = true break time (minutes)
C = mg carbon tetrachloride/liter
A = area of test specimen (cm^2)
F = flow rate (1 liter/minute)

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A.

5.1.1 Level A. The cloth, shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1136, except that each roll of cloth of like description shall be enclosed within two close-fitting clear polyethylene film bags or tubes. The bags or tubes shall be fabricated from polyethylene film having a thickness of 0.003 inch (± 20 percent tolerance). The polyethylene bag shall be formed with heat sealed seams that are straight, continuous, and parallel to each other and the formed edges of the bag. The open end(s) of the bag or tube shall be heat sealed or secured with a mechanical tie (paper or vinyl plastic covered soft steel wire, aluminum band, etc.). Prior to or during the closure operation, excess air within the bag or tubing shall be removed. The first bag or tube around the roll of cloth shall be sealed or secured as specified before insertion within the second bag or tube.

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5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Levels A, B, and Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1136.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1136.

6. NOTES

6.1 Intended use. The type I and III cloths are intended for use in chemical protective overgarments. Type II cloth is intended for use in chemical and fire protective overgarments.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Type required (see 1.2).
c. When a first article is required (see 3.1., 4.3, and 6.3).
d. Selection of applicable level of packing (see 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample consisting of a 3-yard sample of the base cloth, a 3-yard sample of the laminated cloth which has not been impregnated, and a 3-yard sample of the finished laminated cloth. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for inspection and approval of the first article.

6.4 Material impregnation information. Use of the mixtures, and padding described in 6.4.1 through 6.4.2 has been found to produce impregnated cloth that meets the requirements of this document.

6.4.1 Carbon slurry. The following suggested formulation has been found to produce a satisfactory carbon slurry for types I, II, and III material when mixed with other ingredients as required (see 6.4.1.1 thru 6.4.1.3).

Carbon slurry (40% solids) 1/	Percent
Carbon (activated) Grancar 1800	35.0
Gransperse M-60 (18% solids)	5.0
Water	29.0
Grantex 288 (28% solids) 5% solution	25.0
Grantex CP (38% solids)	6.0
	100.0

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1/ Grant Chemical Company, 125 Main Avenue, Elmwood Park, New Jersey 07407.

The carbon should be dispersed in water with the Gransperse M-60 and colloid milled or subjected to similar high shear mixing to obtain a good dispersion of the particles. A 5-percent solution of Grantex 288 should be added to the latex Grantex CP. The thickened latex and dispersed carbon should then be combined.

6.4.1.1 Impregnating mixture for type I cloth:

	Percent
Carbon slurry (see 6.4.1)	100.0

6.4.1.2 Impregnating mixture for type II cloth:

	Percent
Carbon slurry (See 6.4.1)	84.0
Phos Chek^{R} P-30	4.0
Water	12.0
	100.0

Suspend the Phos Chek^{R} P-30 in water and add the suspension to the prepared carbon slurry. Mix thoroughly.

6.4.1.3 Impregnating mixture for typd III cloth:

	Percent
Carbon slurry (see 6.4.1)	70.0
Formulation B 1/	11.0
Water	19.0 2/

1/ See 3.2.3.3 for preparation of formulation B.

2/ The amount of water may be decreased as needed to obtain the laminated cloth weight.

6.4.1.4 Comparable suggested products.

Activated carbon with an average particle size of 15 microns, sources:

a. Calgon Corporation, Pittsburg, PA 15230
b. Carib International, 125 5th Avenue, Paterson, NJ 07508
c. Barnebey Cheney, Cassady at 8th Avenue, Columbus, Ohio 43219

Dispersing agent SMA-1440 Arco Chemical Co., 1500 Market Street, Philadelphia, PA 19101.

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Thickening agent CMC-7M Hercules, Inc., Wilmington, DE 19899

Latex binder Hycar 2671 B.F. Goodrich Chemical Co., 6100 Oak Tree Blvd., Cleveland, Ohio 44131.

6.4.1.5 Sources of prepared slurry. Suggested but not mandatory sources for carbon dispersion are:

Grant Chemical Corp.
125 Main St.
Elmwood Park, NJ 07407

Metro Chemical Co.
325 Valley Avenue
Providence, RI

Trimont Chemicals Inc.
(Protex-a-Cote, Inc.)
27 Haynes Avenue
Newark, NJ 07114

Technical Industries Inc.
P.O. Box 65
Peace Dale, RI 02883

6.4.1.6 Source of supply for mandatory ingredient.

1. Polyox WSRN-80^{R} (For type III)
   Union Carbide Corp.
   Specialty Chemical and Plastic Division
   Old Ridgebuty Road
   Danbury, CT 00817

2. Phos Chek^{R} P-30 (For type II and type III)
   Monsanto Chemical Co.
   800 N. Lindberg Blvd.
   St. Louis, MO 63166

6.4.2 Impregnation padding and drying (types I, II, and III). Padding of the mixture should be accomplished by one pass through squeeze rolls followed by drying at 380°F to attain a final dry add-on necessary to meet the requirements of this document. If a three phase drying oven is used, the settings should be 350°F, 370°F, and 390°F.

6.5 Compressometers. Compressometers that have been found to be satisfactory may be purchased from any one of the following (see 4.5.1.2):

Instron Engineering Corporation
2500 Washington Street
Canton, MA 02021

W.C. Dillon and Company, Inc.
14604 AE Keswick Street
Van Nuys, CA 91405

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Baldwin-Lima-Hamilton Corporation
Industrial Equipment Division
Philadelphia, PA 19142

Tinius-Olsen Testing Machines
Willow Grove, PA 19090

6.6 Warning. Untreated and processed goods should be protected from exposure to chemical vapors, such as solvents, to prevent contamination of the activated carbon.

6.7 Toxicity. Due to the toxicity of carbon tetrachloride, the effluent must be vented into a hood. The exit gases of the vacuum pump and manifold must be vented (see 4.5.2.1.1).

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - GL

Review activities:

Army - MD
DLA - CT

Preparing activity:

Army - GL

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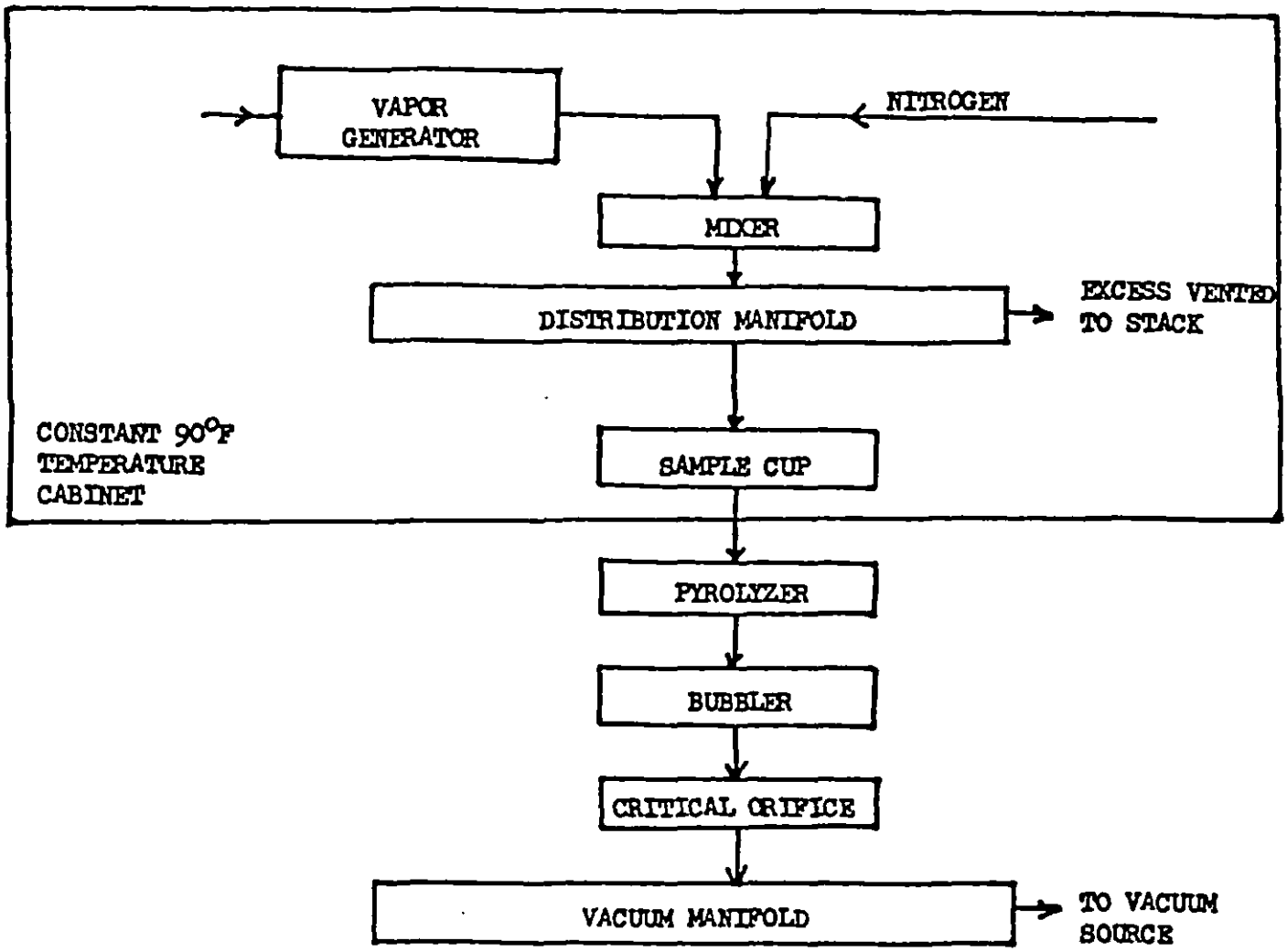
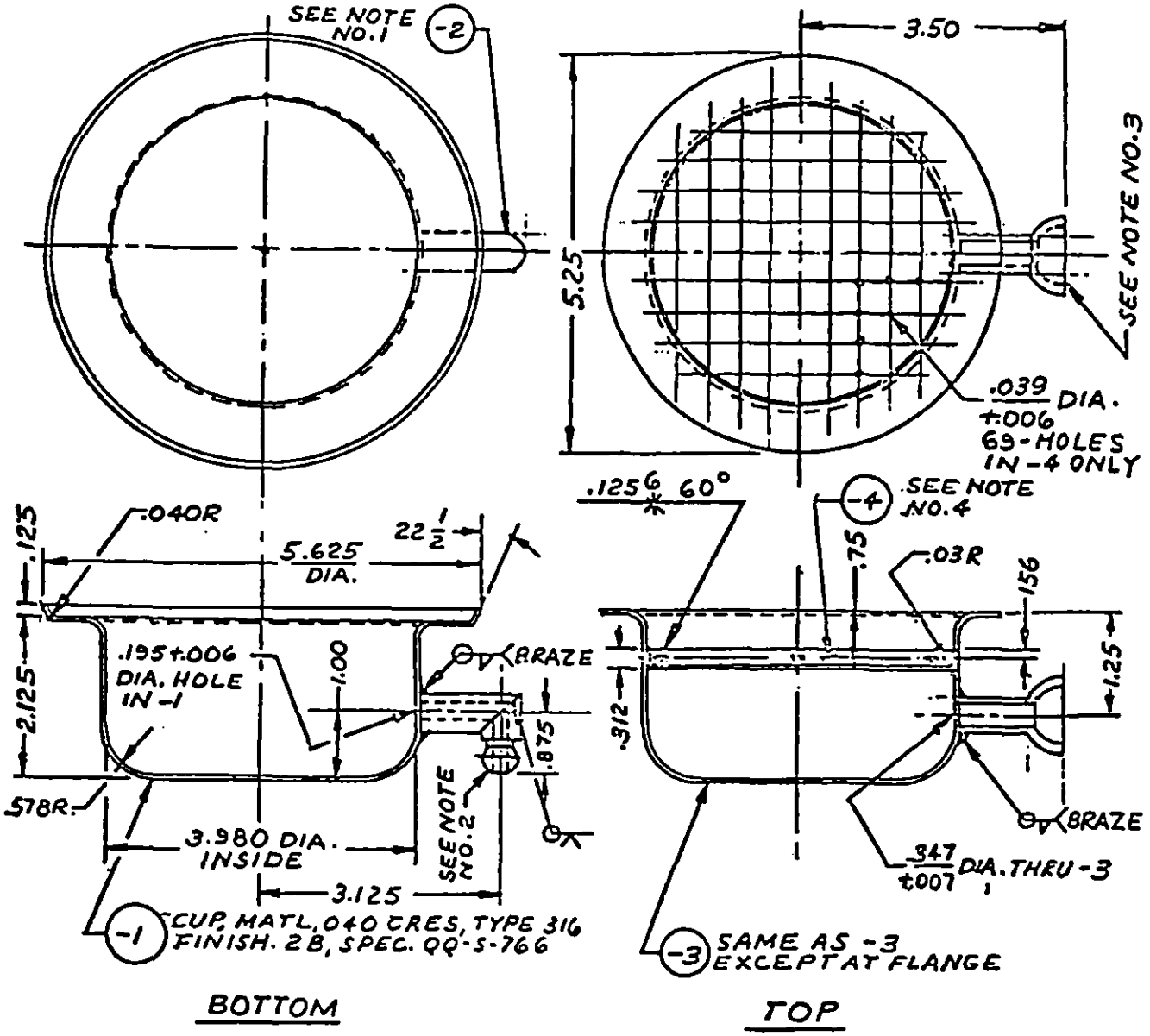


FIGURE 1

FLOW DIAGRAM OF VAPOR PENETRATION APPARATUS

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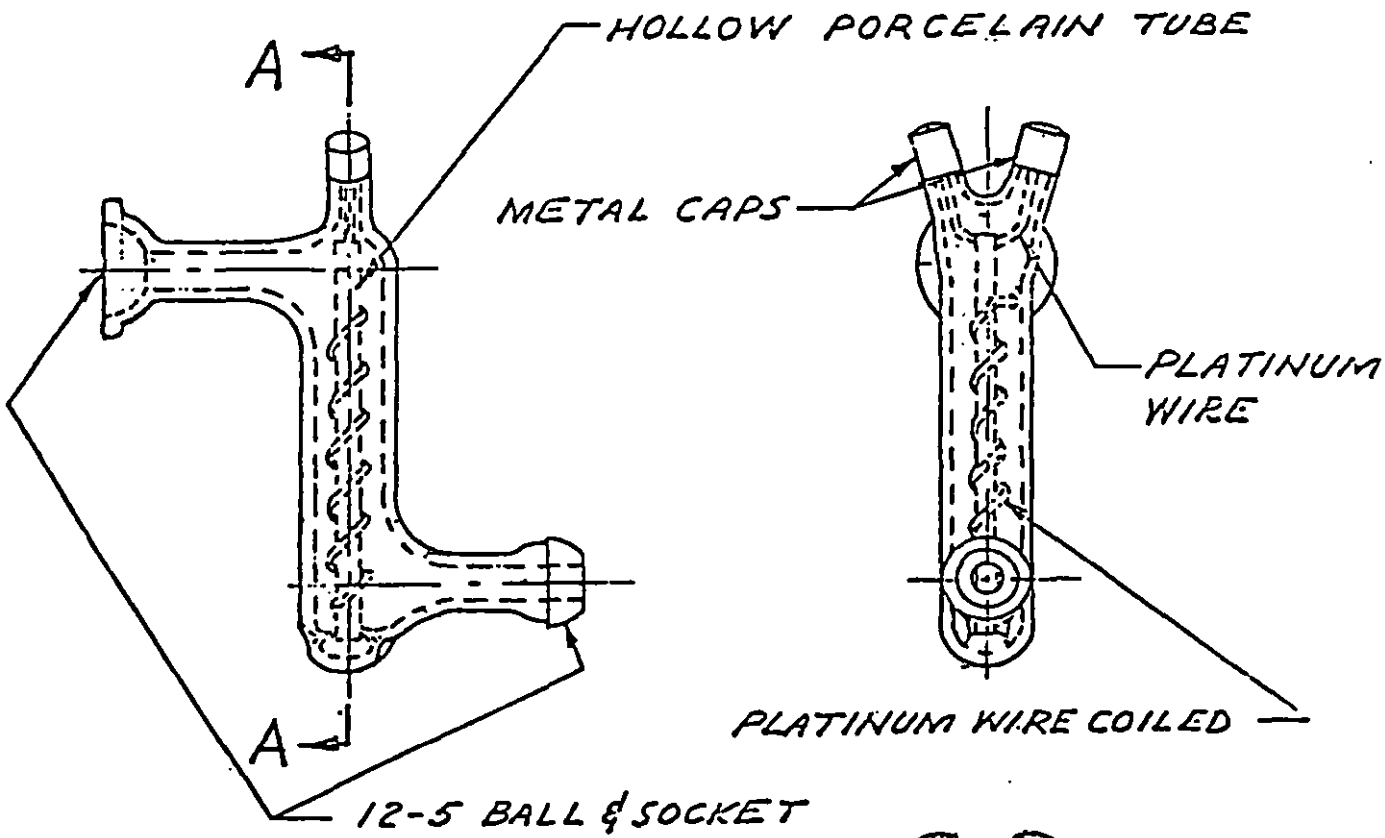


NOTE:

1. 540-.010 O.D. x .190 +.006 I.D. CRES TYPE 316, SPEC. QQ-S-763.
2. GROUND JOINT, S, METAL BALL SIZE 12/5, TYPE 316, CRES. KONTES, TECH GLASS CO., VINELAND, N.J.
3. GROUND JOINT S, METAL SOCKET SIZE 18/9 TYPE 316 CRES. KONTES TECH GLASS CO., VINELAND, N.J.
4. MAT'L., .021 THICK, CRES, TYPE 316, FINISH 2B, SPEC QQ-S-766
5. REMOVE ALL BURRS & BREAK SHARP EDGES.

FIGURE 2. FABRIC SAMPLE CUP

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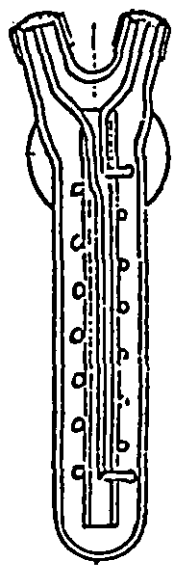


Materials:

(1) 18 gauge platinum wire used in leads.

(2) 30 gauge platinum wire used for heating element. Wire is 36 inches long and is wound around a 0.040 diameter wire to produce a coiled element. The 0.040 diameter wire is removed from the coiled wire. One inch of wire is left uncoiled on each end of the heating element to facilitate fusion to the leads.

(3) The coiled heating element is wrapped around a hollow alundum tube 1/8 inch O.D. and 1/6 inch I.D.



Scale 1/1

SECTION A-A

FIGURE 3
PYROLYZER

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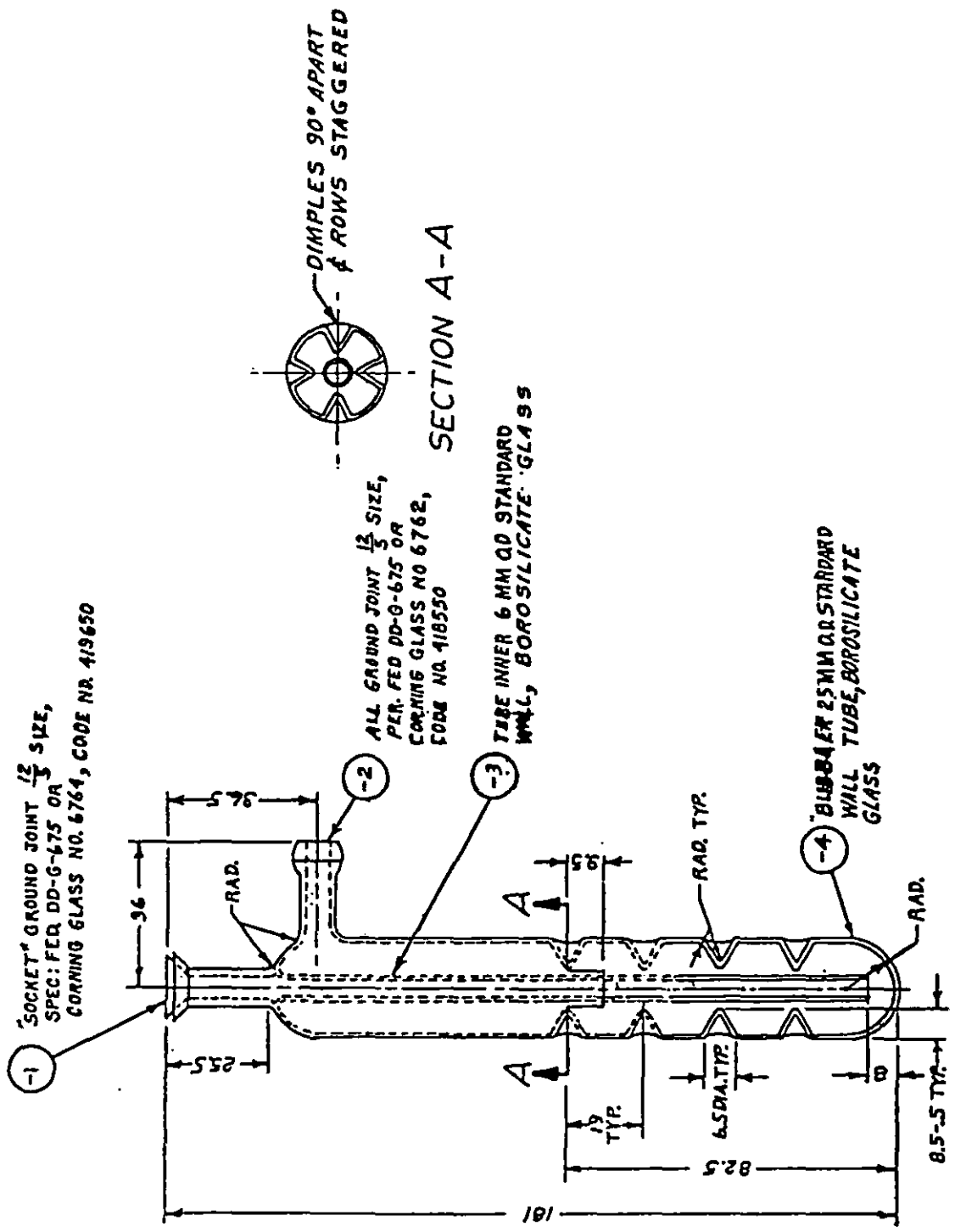


FIGURE 4. BUBBLER

NOTE
1/ ALL DIM. ARE IN MILLIMETERS.
TOL.: ± 1 MM

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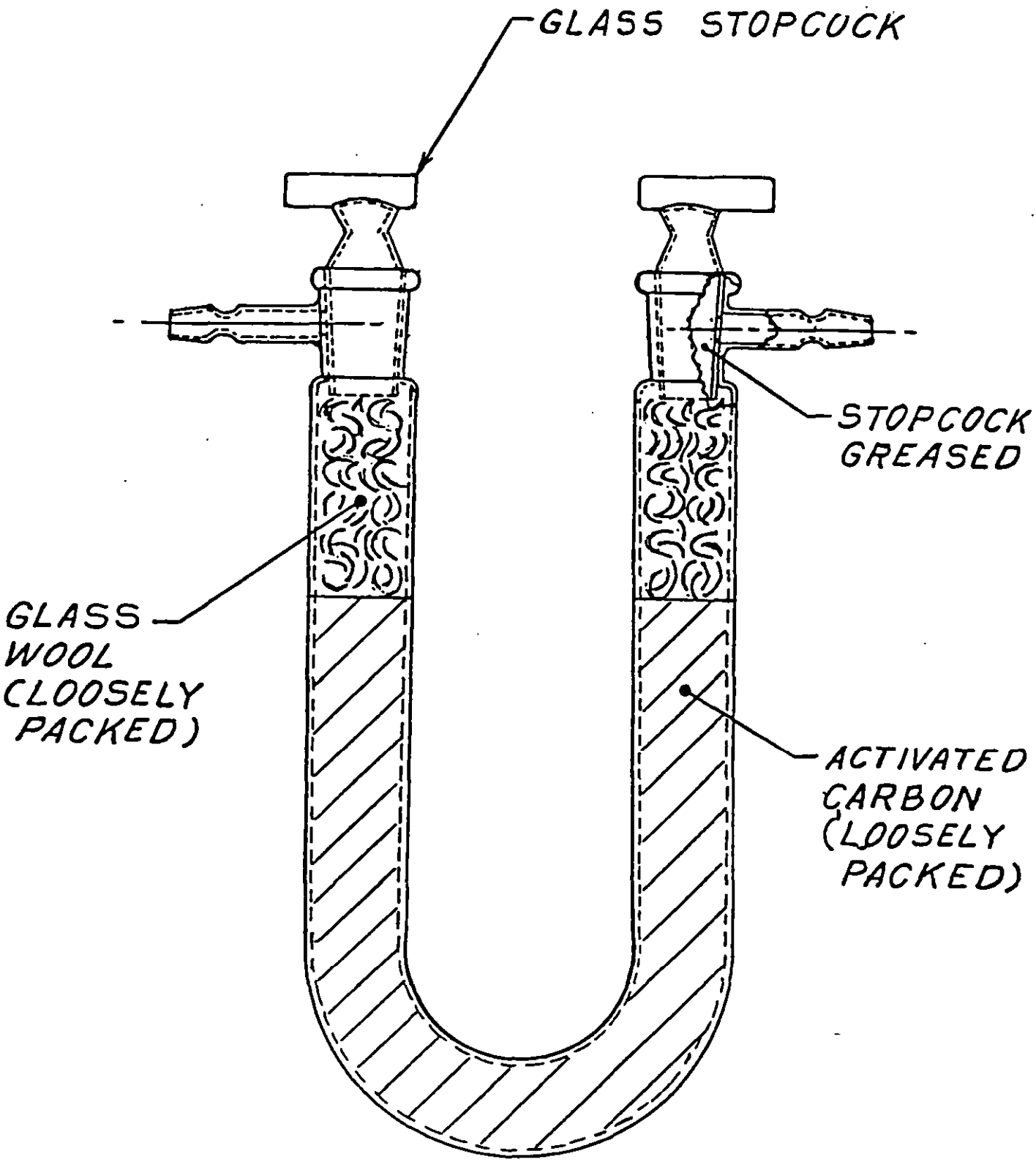


FIGURE 5.

SCHWARTZ TYPE DRYING TUBES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-43858B(GL)	Cloth, Laminated, Nylon Tricot Knit, Polyurethane Foam Laminate, Chemical Protective And Flame Resistant

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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